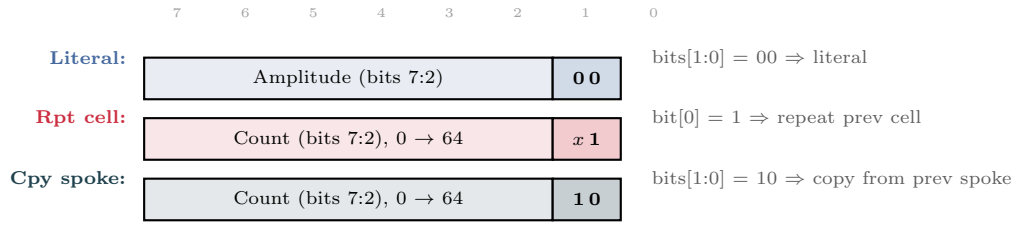
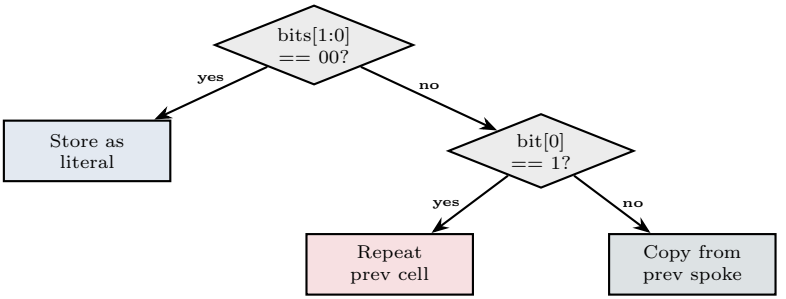
# Encoding 3: 2-Bit RLE with Dual Reference

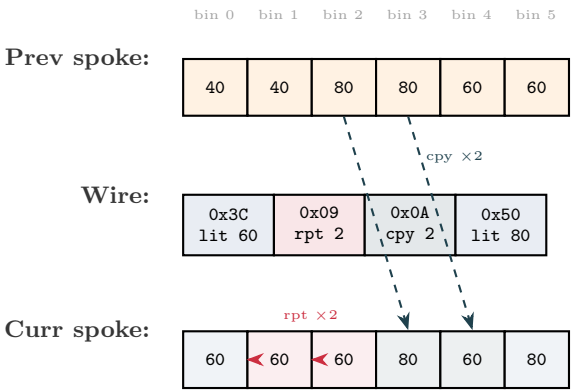
## Byte Layout:




## Decode Logic




## Worked Example




**Key differences from Encoding 1/2:** Control field uses 2 bits (not 1), leaving only 6 count bits. Maximum repeat/copy is 63 (wire) or 64 (when count field is 0). Literal values have bottom 2 bits always zero, so effective amplitudes are multiples of 4 (0, 4, 8, . . . , 252). The `>> 2` display conversion maps these cleanly to 0–63.